

C++ Software Developer Interview Preparation Guide

This PDF contains topic-wise interview preparation material for C++ Software Developer roles. It includes important concepts, brief explanations, and interview-oriented questions and answers.

1. OOP Concepts

Q: What is Polymorphism?

A: Polymorphism allows one interface to behave differently based on the object type. Runtime polymorphism is achieved using virtual functions.

Q: Difference between Overloading and Overriding?

A: Overloading occurs within the same class with different parameters. Overriding occurs in inheritance using virtual functions.

Q: Why use virtual functions?

A: Virtual functions enable runtime polymorphism and dynamic binding.

Q: What is an abstract class?

A: A class containing at least one pure virtual function.

2. Constructors and Destructors

Q: Types of constructors?

A: Default, parameterized, copy, and move constructors.

Q: Why is copy constructor needed?

A: To create a new object as a copy of another object.

Q: Why virtual destructor?

A: Ensures derived class destructor gets called when deleting via base pointer.

Q: What is Rule of 3?

A: If a class defines destructor, copy constructor, or assignment operator, it usually needs all three.

3. Memory Management

Q: Difference between stack and heap memory?

A: Stack memory is automatically managed. Heap memory requires manual allocation/deallocation.

Q: What is memory leak?

A: Memory allocated dynamically but never released.

Q: What is dangling pointer?

A: Pointer pointing to freed memory.

Q: Difference between shallow copy and deep copy?

A: Shallow copy copies pointer values; deep copy duplicates actual memory.

4. STL Concepts

Q: Difference between vector and list?

A: Vector uses contiguous memory and provides fast random access. List is doubly linked and efficient for insertions/deletions.

Q: Difference between map and unordered_map?

A: Map is ordered and implemented using balanced trees. unordered_map uses hashing.

Q: Complexity of vector push_back()?

A: Amortized $O(1)$.

Q: What is iterator?

A: An object used to traverse STL containers.

5. Multithreading

Q: What is race condition?

A: Multiple threads accessing shared data simultaneously causing unpredictable results.

Q: What is mutex?

A: A synchronization mechanism to protect shared resources.

Q: What is deadlock?

A: Two or more threads waiting indefinitely for each other's resources.

6. Linux and Debugging

Q: How do you debug segmentation fault?

A: Using gdb, backtrace, core dump analysis, and memory checking tools.

Q: Useful Linux commands?

A: grep, find, ps, top, chmod, kill, sort.

Q: What is valgrind?

A: A tool used for detecting memory leaks and invalid memory access.

7. C++11/14/17/20 Features

Q: What is auto keyword?

A: Automatically deduces variable type.

Q: What is lambda expression?

A: Anonymous function object used for inline functionality.

Q: What are smart pointers?

A: Pointers that automatically manage memory. Examples: unique_ptr, shared_ptr.

Q: What is move semantics?

A: Transfers ownership of resources instead of copying.

8. Frequently Asked Coding Questions

Q: Reverse a string

A: Use two-pointer approach swapping characters from both ends.

Q: Check palindrome

A: Compare characters from beginning and end until middle.

Q: Reverse linked list

A: Iteratively reverse pointers using prev, curr, next.

Q: Find duplicate in array

A: Use hashing or sorting approach.

9. HR Interview Questions

Q: Tell me about yourself

A: Briefly explain experience, skills, projects, and career goals.

Q: Why do you want to change company?

A: Focus on growth, learning, and better opportunities.

Q: What are your strengths?

A: Problem-solving, debugging, teamwork, ownership.

Q: Describe a challenging bug you solved

A: Explain issue, debugging process, and final solution.

High Priority Revision Topics

Topic	Must Know
OOP	Virtual functions, inheritance, polymorphism
Memory	Leaks, stack vs heap, deep copy
STL	vector, map, iterators
Threading	mutex, deadlock, race condition
Linux	gdb, valgrind, commands

Final Interview Tips:

1. Explain answers clearly and confidently.
2. Focus on debugging and practical experience.
3. Revise projects thoroughly.
4. Practice writing small programs without IDE help.
5. Stay calm and communicate your thought process.